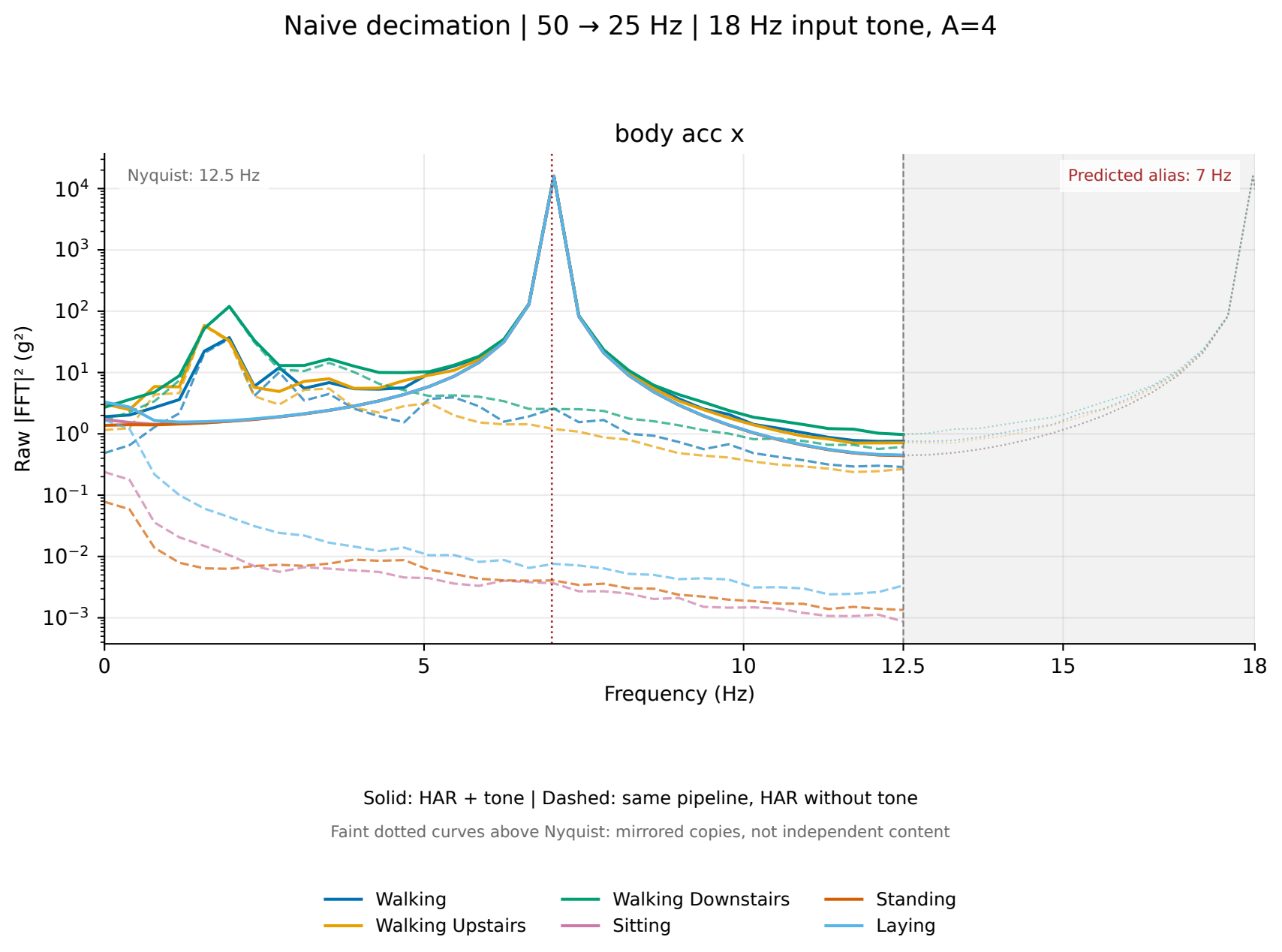
Naive decimation | 50 → 25 Hz | 18 Hz input tone, A=4
body acc x
Nyquist: 12.5 Hz
Predicted alias: 7 Hz
Raw |FFT|² (g²)
Frequency (Hz)
0
5
10
12.5
15
18
Solid: HAR + tone | Dashed: same pipeline, HAR without tone
Faint dotted curves above Nyquist: mirrored copies, not independent content
Walking
Walking Upstairs
Walking Downstairs
Sitting
Standing
Laying